Data: `audi.csv` (Kaggle UK used cars), Audi A3 only. Put `audi.csv` in the same folder as this notebook.

In [3]:
```
# install.packages("ISLR")
library(ISLR)   # loaded per the assignment; lm() and anova() are base R

cars <- read.csv("Data/audi.csv")
cars$model <- trimws(cars$model)      # model names have a leading space

a3 <- subset(cars, model == "A3")     # hold the car model fixed
set.seed(123)
d <- a3[sample(nrow(a3), 50), ]       # sample size: match your original
project

d$Age   <- 2020 - d$year              # dataset was scraped in 2020
d$Price <- d$price / 1000             # in 1,000s (British pounds)
d$Miles <- d$mileage / 1000           # in 1,000s of miles
d <- d[, c("Price", "Age", "Miles")]
head(d)
```

A data.frame: 6 × 3

	Price	Age	Miles
	<dbl>	<dbl>	<dbl>
2691	13.498	6	21.238
3017	30.777	0	3.925
1185	27.881	0	5.432
3393	11.995	5	28.527
1302	12.498	3	37.882
5820	8.990	7	75.142

In [4]:
```r
# (a)
m1 ← lm(Price ~ Age + Miles, data = d)
summary(m1)
anova(m1)
```

```
Call:
lm(formula = Price ~ Age + Miles, data = d)

Residuals:
    Min      1Q  Median      3Q     Max
-5.4568 -2.4726 -0.5328  1.3778 14.1137

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 24.96032    0.90527  27.572  < 2e-16 ***
Age         -1.55795    0.33304  -4.678 2.48e-05 ***
Miles       -0.06155    0.03465  -1.777   0.0821 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.91 on 47 degrees of freedom
Multiple R-squared:  0.6555,    Adjusted R-squared:  0.6408
F-statistic: 44.71 on 2 and 47 DF,  p-value: 1.331e-11
```

A anova: 3 × 5

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
	<int>	<dbl>	<dbl>	<dbl>	<dbl>
Age	1	1318.98355	1318.98355	86.272324	3.282609e-12
Miles	1	48.25224	48.25224	3.156091	8.211701e-02
Residuals	47	718.56447	15.28861	NA	NA

In [5]:
```r
# (b) residual for the first car
b    ← coef(m1)
pred ← b[1] + b[2]*d$Age[1] + b[3]*d$Miles[1]
res  ← d$Price[1] - pred
cat("Predicted:", pred, " Actual:", d$Price[1], " Residual:", res, "\n")
resid(m1)[1]   # check: should match
```

```
Predicted: 14.30541  Actual: 13.498  Residual: -0.8074115
```

2691: -0.807411545296432

In [6]:
```r
# (c) supporting info: correlation between predictors
cor(d)
```

A matrix: 3 × 3 of type dbl

	Price	Age	Miles
Price	1.0000000	-0.7952127	-0.7036298
Age	-0.7952127	1.0000000	0.7606747
Miles	-0.7036298	0.7606747	1.0000000

(c) Use the t-test rows in `summary(m1)` for Age and Miles (hypotheses, t, p-value, conclusion in context).

(d) Use the overall F-statistic line at the bottom of `summary(m1)` (H0: β1 = β2 = 0).

In [7]:
```r
# (a) quadratic fit + scatterplot
m2 <- lm(Price ~ Age + I(Age^2), data = d)
summary(m2)

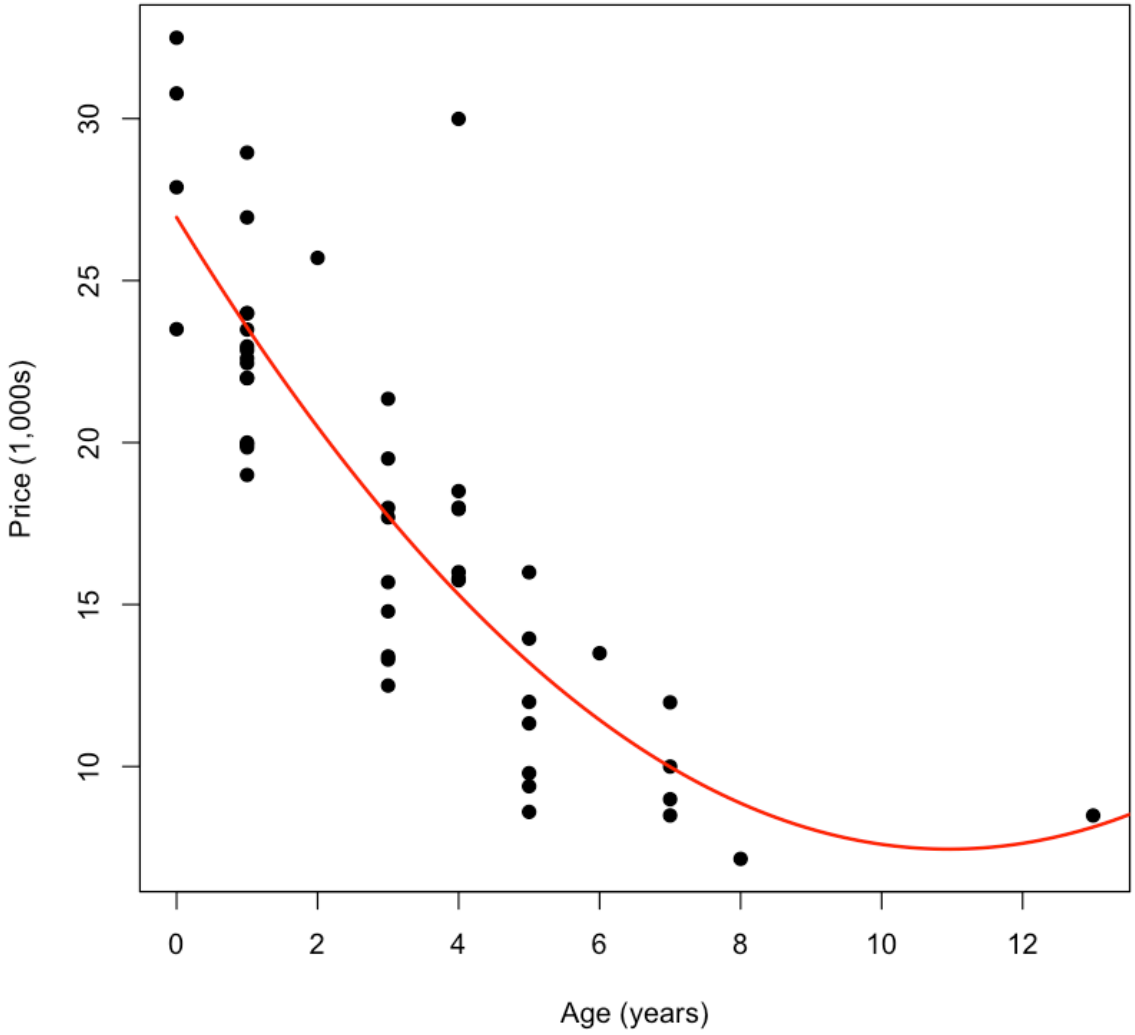
plot(d$Age, d$Price, pch = 19, xlab = "Age (years)", ylab = "Price
(1,000s)")
ages <- seq(min(d$Age), max(d$Age) + 10, length.out = 200)
lines(ages, predict(m2, data.frame(Age = ages)), col = "red", lwd = 2)
abline(h = 0, lty = 2)
```

```
Call:
lm(formula = Price ~ Age + I(Age^2), data = d)

Residuals:
    Min      1Q  Median      3Q     Max
-5.2336 -2.0000 -0.0524  1.9364 14.6828

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 26.95325    1.03917  25.937  < 2e-16 ***
Age         -3.56170    0.48175  -7.393 2.08e-09 ***
I(Age^2)     0.16261    0.04594   3.539 0.000916 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.589 on 47 degrees of freedom
Multiple R-squared:  0.7097,    Adjusted R-squared:  0.6974
F-statistic: 57.46 on 2 and 47 DF,  p-value: 2.376e-13
```


Price (1,000s)
10
15
20
25
30
0
2
4
6
8
10
12
Age (years)

In [8]:
```
# (b) does predicted price reach zero or go negative?
b2 ← coef(m2)
vertex ← -b2[2] / (2*b2[3])
minval ← b2[1] + b2[2]*vertex + b2[3]*vertex^2
disc   ← b2[2]^2 - 4*b2[3]*b2[1]
vertex; minval; disc
polyroot(c(b2[1], b2[2], b2[3]))   # ages where predicted price = 0 (if
real)
range(d$Age)                       # is the vertex inside the observed
range?
```

Age: 10.951607760757

(Intercept): 7.45005689902805

Age: -4.84584504923264

1. 10.9516077607571+6.76869310996754i
2. 10.951607760757-6.76869310996754i

1. 0
2. 13

In [9]:
```
# (d) cubic term
m3 ← lm(Price ~ Age + I(Age^2) + I(Age^3), data = d)
summary(m3)
anova(m2, m3)
```

```
Call:
lm(formula = Price ~ Age + I(Age^2) + I(Age^3), data = d)

Residuals:
    Min      1Q  Median      3Q     Max
-5.2375 -2.0028 -0.0544  1.9407 14.6803

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 26.9470226  1.3401438  20.108  < 2e-16 ***
Age         -3.5534284  1.2085497  -2.940  0.00512 **
I(Age^2)     0.1606510  0.2660558   0.604  0.54893
I(Age^3)     0.0001088  0.0145469   0.007  0.99406
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.628 on 46 degrees of freedom
Multiple R-squared:  0.7097,    Adjusted R-squared:  0.6908
F-statistic: 37.49 on 3 and 46 DF,  p-value: 2.061e-12
```

A anova: 2 × 6

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	47	605.4486	NA	NA	NA	NA
2	46	605.4478	1	0.0007367564	5.597641e-05	0.9940628

(a) Price = β0 + β1·Age + β2·Miles + β3·Age² + β4·Miles² + β5·(Age·Miles) + ε

In [10]:
```
# (b)
m4 <- lm(Price ~ Age + Miles + I(Age^2) + I(Miles^2) + Age:Miles, data = d)
summary(m4)
anova(m4)
coef(m4)
```

```
Call:
lm(formula = Price ~ Age + Miles + I(Age^2) + I(Miles^2) + Age:Miles,
    data = d)

Residuals:
    Min      1Q  Median      3Q     Max
-6.7935 -2.1337 -0.2836  1.6668 13.6954

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 26.2076179  1.1026423  23.768  < 2e-16 ***
Age         -4.5165195  0.9115460  -4.955 1.12e-05 ***
Miles        0.2126182  0.1323215   1.607    0.115
I(Age^2)     0.3407773  0.1584301   2.151    0.037 *
I(Miles^2)  -0.0008750  0.0009857  -0.888    0.379
Age:Miles   -0.0284408  0.0300782  -0.946    0.350
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.543 on 44 degrees of freedom
Multiple R-squared:  0.7352,    Adjusted R-squared:  0.7052
F-statistic: 24.44 on 5 and 44 DF,  p-value: 1.093e-11
```

A anova: 6 × 5

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
	<int>	<dbl>	<dbl>	<dbl>	<dbl>
Age	1	1318.98355	1318.98355	105.091967	3.088756e-13
Miles	1	48.25224	48.25224	3.844568	5.625675e-02
I(Age^2)	1	120.62111	120.62111	9.610665	3.369054e-03
I(Miles^2)	1	34.48863	34.48863	2.747933	1.044943e-01
Age:Miles	1	11.22150	11.22150	0.894090	3.495353e-01
Residuals	44	552.23323	12.55076	NA	NA

(Intercept)
: 26.2076179005369

Age
: -4.51651950425601

Miles
: 0.212618229734629

I(Age^2)
: 0.340777330225797

I(Miles^2)
: -0.000875048408654596

Age:Miles
-0.0284408313855714

In [11]:
```r
# (c) nested F-test: drop Miles, Miles^2, Age*Miles (3 terms)
m_red ← lm(Price ~ Age + I(Age^2), data = d)
anova(m_red, m4)

SSE_R ← sum(resid(m_red)^2);  df_R ← df.residual(m_red)
SSE_F ← sum(resid(m4)^2);     df_F ← df.residual(m4)
k      ← df_R - df_F
F_stat ← ((SSE_R - SSE_F) / k) / (SSE_F / df_F)
p_val  ← pf(F_stat, k, df_F, lower.tail = FALSE)
c(SSE_R = SSE_R, SSE_F = SSE_F, k = k, df_F = df_F, F = F_stat, p = p_val)
```

A anova: 2 × 6

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	47	605.4486	NA	NA	NA	NA
2	44	552.2332	3	53.21532	1.413337	0.2515768

SSE_R
605.448557200011
SSE_F
552.233232485336
k
3
df_F
44
F
1.41333658178438
p
0.251576816896111

In [12]:
```
models ← list(Linear_Age_Miles = m1, Quadratic_Age = m2, Cubic_Age = m3,
Full_2nd_Order = m4)
data.frame(
  adjR2 = sapply(models, function(m) summary(m)$adj.r.squared),
  s     = sapply(models, function(m) summary(m)$sigma),
  AIC   = sapply(models, AIC),
  BIC   = sapply(models, BIC)
)
```

A data.frame: 4 × 4

	adjR2	s	AIC	BIC
	<dbl>	<dbl>	<dbl>	<dbl>
Linear_Age_Miles	0.6408373	3.910065	283.1555	290.8036
Quadratic_Age	0.6973764	3.589134	274.5912	282.2393
Cubic_Age	0.6907980	3.627935	276.5911	286.1512
Full_2nd_Order	0.7051554	3.542705	275.9912	289.3754

In [13]:
```
# residual plots for your recommended model (change m4 as needed)
par(mfrow = c(2, 2)); plot(m4); par(mfrow = c(1, 1))
```

